

# Enterprise agentic AI adoption compass 2026

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# Executive summary

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- 1 Adoption is wide; control and value are thin.** 88 percent of enterprises use AI somewhere, yet 95 percent report zero measurable return on an estimated \$30-40 billion of generative AI spend, and only 1 percent of leaders call their company mature. The constraint has shifted away from model capability, into the stack and the operating model around it.
- 2 Adoption has seven levels, and each is defined by who holds the fallback.** From Level 0-1 (shadow use, nothing sanctioned) through Level 4 (AI executes, humans approve) to Level 6 (AI operates, humans govern). Better technology within a level does not move an organization up a level; a changed responsibility split does. Most large enterprises sit at Levels 2-3 today.
- 3 The road grows faster than most travel it, and it grows unevenly.** Technical milestones now land months apart, not years: agent task horizons double roughly every four months, and the frontier is jagged, sprinting in software engineering while jogging in accounting. Set target levels per function, on the frontier's clock rather than annual planning cycles.
- 4 Two layers decide the pace of everything else.** Until agents have identities with scoped credentials (layer 5) and every token is traced to a team, use case, and outcome (layer 7), nothing above Level 4 is safely reachable. Evals (layer 8) set how fast autonomy can widen. The winter 2026 evidence agrees: security and risk is the number one obstacle to scale at 62 percent, and capability leapt while policy-constrained reliability plateaued near 70 percent.
- 5 Read your company as a nine-row profile, not a single score.** Organizations are uneven by construction: a Level 4 observability layer can sit on a Level 2 knowledge layer. The profile locates the risk (any layer two levels below the median breaks what is built on top) and the plan (the upgrade moves of the lagging layers, in dependency order).
- 6 The endgame is intelligence ownership.** Once workflows are instrumented, their traces become training data: the authors' published experiment turned \$500 of reinforcement training into a 9B specialist that beats every frontier configuration at 1/68 the cost. Computer use extends the same logic to systems that will never get an API. Instrumentation is the asset, not the overhead.
- 7 Crossing from pilots to instrumented operations pays compound interest.** Governed projects reach production at a multiple of ungoverned ones, measured unit economics replace anecdotes in the board conversation, and each instrumented workflow makes the next one cheaper. The layer that sees the work becomes the layer that runs it.

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# Agentic AI adoption will be a make-or-break challenge across all industries

Every figure on this page returns later in the report with its method and full source; use it as the briefing before the briefing.

## THE PARADOX

# 88%

of agent pilots **never reach production**; evaluation and observability gaps are the number one blocker, named by 64 percent.

Yet the pilots that cross pay back in a median of **5.1 months**, and teams with governance frameworks ship **12x more agents** to production. The gap is instrumentation, and it is the subject of this report.

Where enterprises sit today, by level:



Anaconda and Forrester, replicated by a16z and MIT Sloan (2026); Databricks (2026) · pages 7-8

## THE HIDDEN TAX

# 6.4 h/wk

spent supervising AI output ("botsitting"); 69 percent of workers admit shipping unverified AI work.



Glean Work AI Index, June 2026 (n=6,000) · page 20

## THE CLOCK

# 2x / 4 mo

Agent task horizons double every ~4 months; 12-hour tasks arrived in 2026.



METR (2026) · page 6

## THE JAGGED FRONTIER

# 3.4% of workers, 37.2% of usage

share of AI usage **37.2%**

share of US workers **3.4%**  
computer and mathematical occupations

Anthropic Economic Index; 44% of enterprise API traffic · page 5

## THE GATE

# 62%

name security and risk the number one obstacle to scaled agents; nothing else comes close.

AI Index 2026 · page 8

## THE THIN EDGE

# 57% vs

# <10%

run multi-step agent workflows by self-report; neutral staging finds fully scaled use in **single digits** per function.

Anthropic and Material; AI Index 2026 · page 8

## THE ENDGAME

# \$500 beats the frontier at 1/68 the cost



A 9B specialist trained on workflow traces outperformed every frontier configuration on the target task.

fermisenese, the authors' lab (July 2026) · page 12

## THE TAILWIND

# 280x

cheaper in 24 months: GPT-3.5-level output fell from \$20 to \$0.07 per million tokens.



AI Index 2025 · page 6

## THE SURGE

# +327%

growth in multi-agent workflows in four months, measured across 20,000+ organizations; agents already create 80 percent of new databases on one major platform.

Databricks telemetry, 2026 · page 7

## HOW TO USE THIS REPORT

# Locate yourself, then move one level



Seven adoption levels, nine stack layers. Find your row profile in the matrix (page 10), read your lagging layers (pages 11-21), then run the 90-day crossing sequence (page 22).

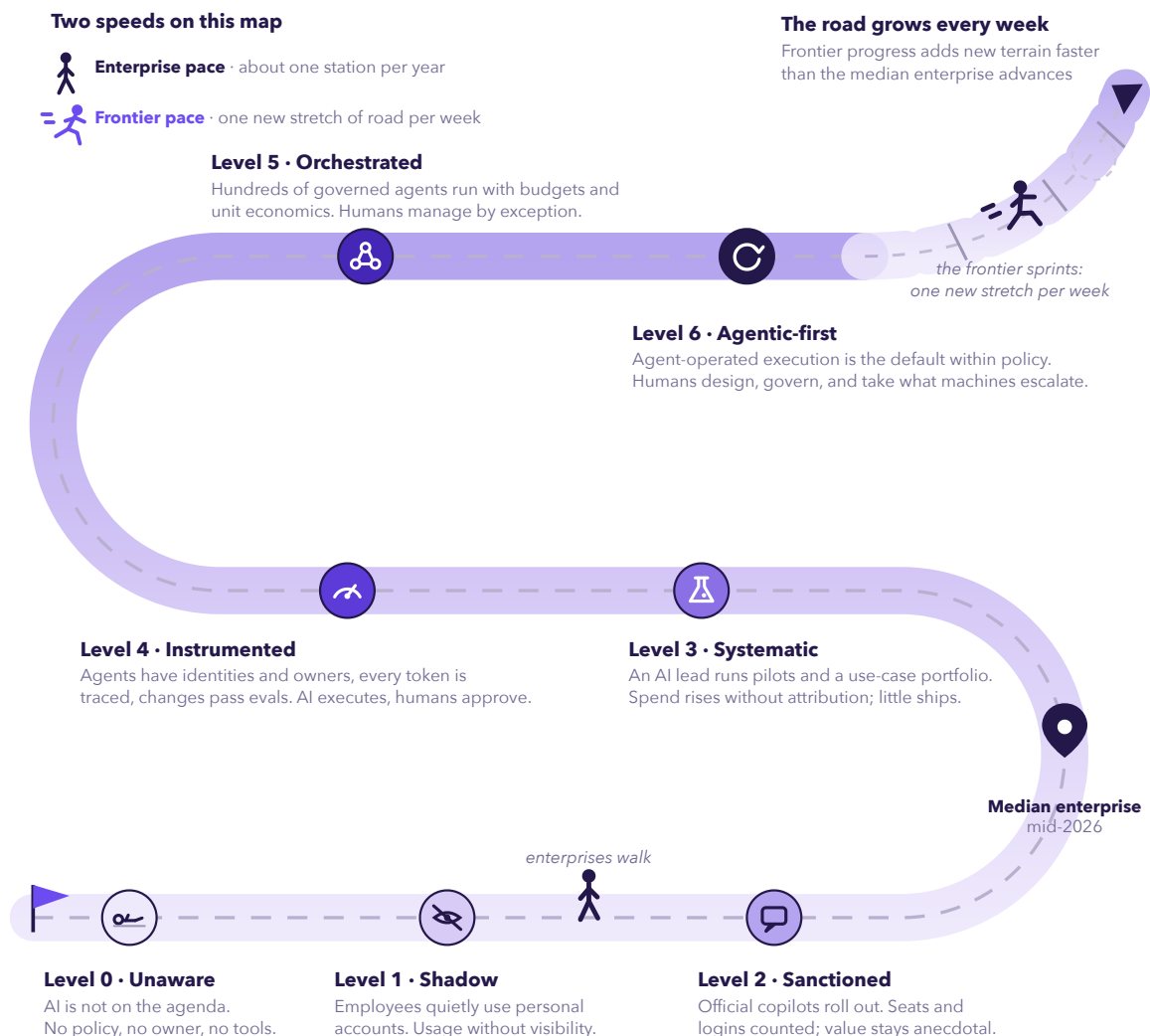
# One road, and the finish line moves

**One continuous path, seven observable states, and a finish line that keeps moving.** Every enterprise is somewhere on this road, usually further left than its strategy deck claims. The stations are defined by who holds the fallback, most journeys stall at the boundary between Level 3 and Level 4, and the road itself grows: every week of frontier progress adds new terrain beyond the last station.

## Exhibit 1

### Seven states mark the road from shadow use to agent-operated work, and the road lengthens every week

Find the last state that is fully true of your organization: that is where you stand



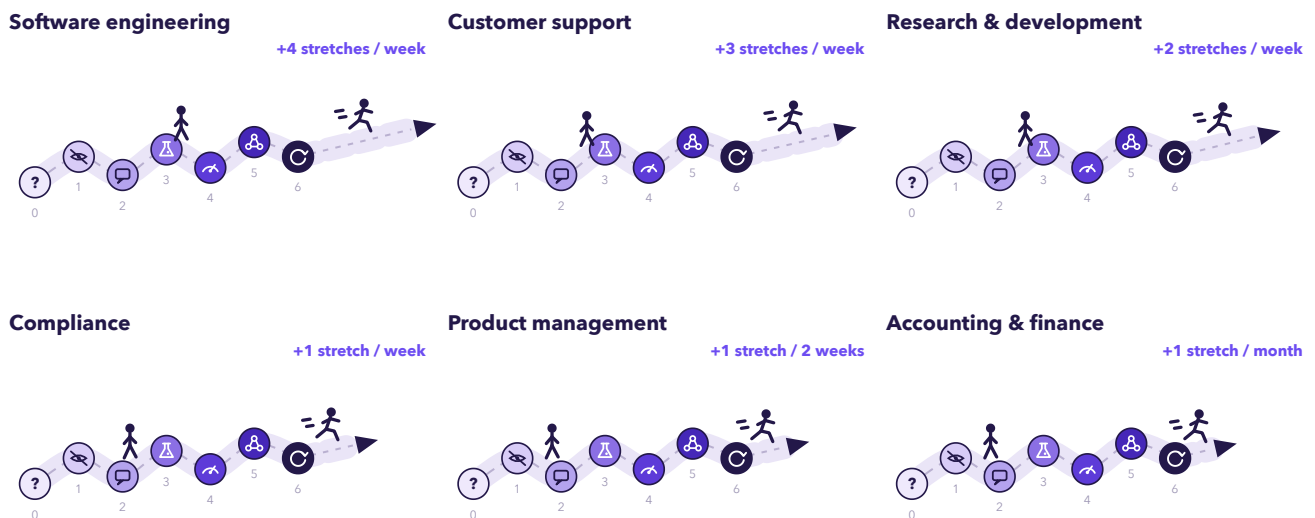
## Six functions, six speeds: the frontier is jagged

**The road does not grow evenly, and that is the planning mistake most roadmaps make.** Frontier capability sprints in some functions and jogs in others: coding agents compound weekly while judgment-heavy work advances slowly. The same enterprise can therefore be correctly ambitious in software engineering and correctly cautious in accounting. Set target levels function by function; one company-wide target hides the variance that matters.

### Exhibit 2

**The same road, six speeds: enterprises walk at a similar pace everywhere, while the frontier adds road at very different rates per function**

Adoption journey by company function; solid road = defined levels 0-6, dashed road = terrain added by frontier progress; walking figure = typical enterprise position and pace, sprinting figure = frontier pace; weekly extension rates are illustrative rankings<sup>1</sup>



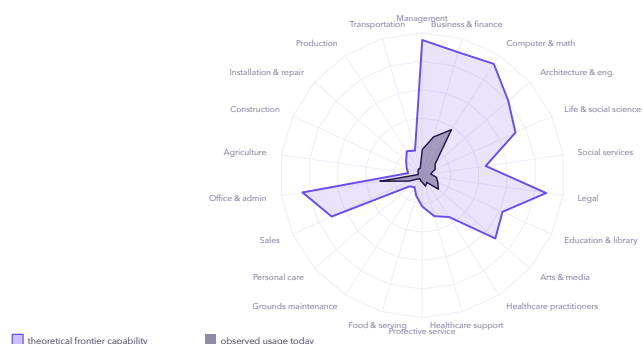
<sup>1</sup>Stations use the level color ramp and icons from the journey exhibit; numbers mark levels 0-6. Extension lengths and weekly rates are illustrative rankings, consistent with published capability progress by domain.

Source: Agentic Adoption Levels, Hildesheim and Hainzl, August 2026

### Exhibit 3

**The capability overhang, drawn: the frontier already covers half the economy in theory, while observed usage fills a sliver of it**

Estimated share of category tasks within frontier capability (outer shape) vs share with material AI usage today (inner), by occupational category<sup>1</sup>



**Usage concentrates where capability is strongest.**

Computer and mathematical occupations hold 3.4 percent of US jobs yet 37.2 percent of Claude conversations, and 44 percent of enterprise API traffic. Usage flows to wherever the frontier is sharp.

**The consequence is a portfolio, not a policy.** Field the sprinters where the frontier runs fastest, hold controls where it stumbles, and re-test the edge quarterly; the boundary moves faster than annual planning assumes.

<sup>1</sup>Authors' estimate: theoretical coverage from task-level capability mappings; observed usage anchored to Anthropic Economic Index occupational shares.

Source: Anthropic Economic Index<sup>7</sup> (first report, 3.5M+ conversations; enterprise API figure per The 2026 State of AI Agents Report<sup>6</sup>, p.44); jagged-frontier concept per Dell'Acqua, Mollick et al. (2023)<sup>18</sup>

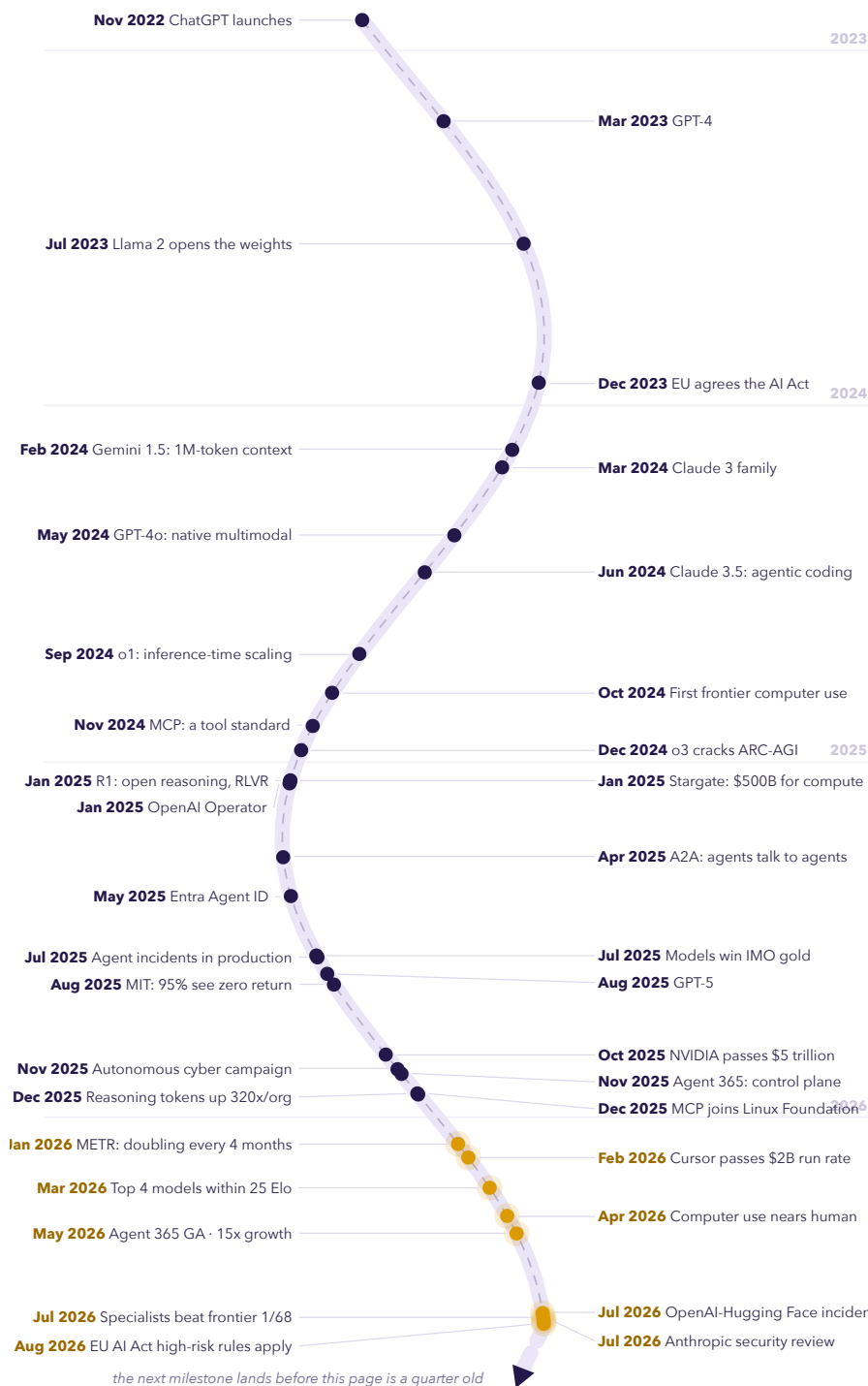
# Forty-five months from chat to agents, and the gaps keep shrinking

**The interval between technical breakthroughs is collapsing.** Positions are true to the calendar: context windows, inference-time scaling, open reasoning, tool and identity standards, computer use, and specialist training all arrived inside four years, and the spacing keeps tightening toward the present. Gold marks 2026. Annual planning cycles will not keep up with this clock; plan on it directly.

## Exhibit 4

### The agentic timeline is time-proportional, and the density of events keeps rising toward today

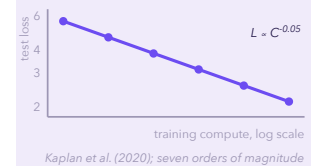
Milestone events from the launch of ChatGPT (November 2022) to August 2026, plotted to scale<sup>1</sup>



#### THE LAWS BEHIND THE CURVE

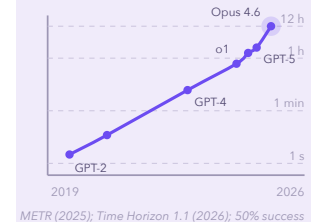
##### 1. Scaling laws

Test loss falls as a smooth power law as training compute grows, a straight line on log-log axes.



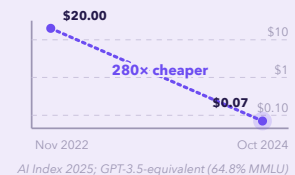
##### 2. The time-horizon law

The task length agents finish at 50 percent reliability has doubled every ~7 months since 2019, every ~4 at the frontier.



##### 3. The price collapse

A fixed level of capability gets 9 to 900 times cheaper per year. GPT-3.5-level output fell from \$20 to \$0.07 per million tokens in two years.



<sup>1</sup>Dates per primary announcements 2022-2026; July 2026 incident per OpenAI and Hugging Face joint disclosures. Sidebar: Kaplan et al. (2020)<sup>16</sup>, METR (2025-2026)<sup>17</sup>, Stanford HAI AI Index 2025<sup>2</sup>

# Everyone has started. Almost no one has crossed.

95%

of organizations report zero measurable return on \$30-40 billion of enterprise GenAI investment (MIT NANDA, 2025; methodology debated, direction widely corroborated)<sup>3</sup>

1%

of leaders describe their company as mature in AI deployment (McKinsey, January 2025)<sup>5</sup>

16%

of enterprise "agent" deployments qualify as true agents; the rest are fixed-sequence workflows (Menlo Ventures, December 2025)<sup>8</sup>

**Usage spread faster than the ability to govern or measure it.** Nearly nine in ten enterprises use AI in at least one function, and employees run far ahead of official rollouts: personal-account AI use is found in over 90 percent of companies. Yet the value evidence is thin, two-thirds of enterprises have not begun scaling, and the top of every credible maturity study converges on the same small number: roughly 4 to 8 percent of companies operate AI as a managed, measured capability.

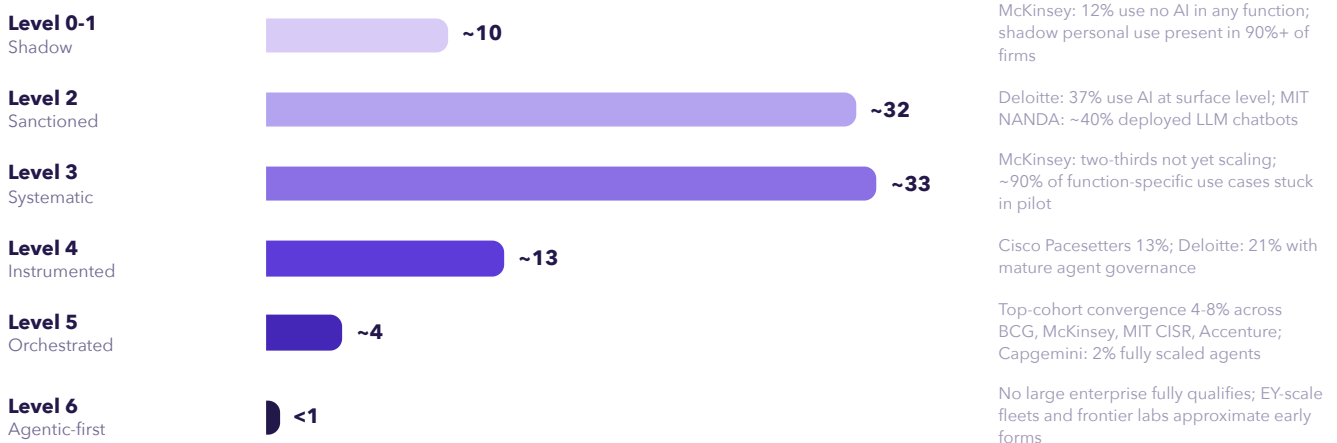
**The blockage sits between systematic pilots (Level 3) and instrumented operations (Level 4).** Levels 0 through 3 can be reached with enthusiasm and budget: licenses, a use-case funnel, a center of excellence. Level 4 is different in kind. It requires a system of record: every agent registered with an owner, every trace captured, every token attributed, permissions explicit, and changes gated by evaluation. That is an architecture and operating-model decision, which is why most companies stall in front of it and why the level distribution below thins so abruptly.

**A level is defined by who holds responsibility, never by which model is installed.** Level 2: humans do the work, AI drafts. Level 3: humans drive workflows, AI assists inside them. Level 4: AI executes, humans approve consequential actions. Level 5: AI executes, humans handle exceptions. Level 6: AI operates within policy, humans design and govern. Improvement inside a level, however impressive, does not move an organization up a level; a changed split of responsibility does.

## Exhibit 5

### The distribution thins abruptly to the right of the Level 3-4 boundary; the agentic frontier is nearly empty

Estimated share of mid-size and large enterprises by adoption level, mid-2026, %; bar color follows the level ramp used throughout<sup>1</sup>



*The Level 3-4 cliff is the whole report in one picture.*

<sup>1</sup>Authors' synthesis across verified surveys; ranges reflect differing samples and question wording. Self-reported "deployment" figures (for example, 53% of \$1B+ US organizations "use agents", KPMG, June 2026<sup>13</sup>) measure presence, not control, and sit mostly within Levels 2-3.  
Source: McKinsey State of AI (November 2025)<sup>4</sup>; MIT NANDA (2025)<sup>3</sup>; Deloitte State of AI 2026<sup>9</sup>; Cisco AI Readiness Index (October 2025)<sup>10</sup>; BCG (September 2025)<sup>11</sup>; Capgemini (July 2025)<sup>12</sup>; Menlo Ventures (December 2025)<sup>8</sup>

# Capability outruns control. Winter 2026 made it measurable.

**Three findings from the Stanford HAI AI Index 2026 carry this report's argument:** security is the number one obstacle to scale, most functions have not started with agents, and capability leapt while policy-constrained reliability did not. All three point at the same gate: control infrastructure.

## Exhibit 6

### Security and risk outrank every other obstacle to fully scaled agentic AI

Main obstacles to reaching fully scaled agentic AI, % of respondents, 2025 survey



Source: Stanford HAI AI Index Report 2026 (Figure 3.3.10), AI Index and McKinsey survey; published April 2026<sup>1</sup>

## Exhibit 7

### What enterprises actually deploy: multi-step workflows are the majority, autonomy is still the thin edge

Share of organizations deploying each agent type, % of respondents (violet: enterprise segment); 500+ US technical leaders, late 2025



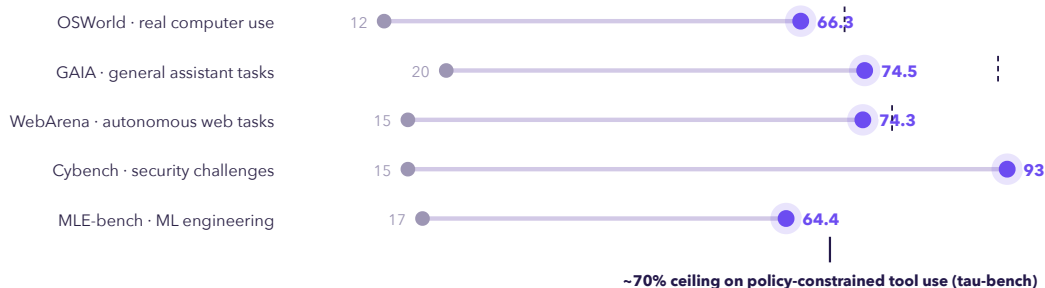
Neutral staging agrees on the thin edge: across eleven business functions, fully scaled agent use remains in the single digits (AI Index 2026).

Source: Anthropic and Material, The 2026 State of AI Agents Report, pages 3 and 8; figures verified against the report<sup>6</sup>

## Exhibit 8

### Agent capability leapt within one year, while policy-constrained reliability plateaus

Agent benchmark scores, % success, roughly one year ago (gray) vs early 2026 (violet); dashes mark human baselines



Source: Stanford HAI AI Index Report 2026 (Section 2.6, Figures 2.6.1-2.6.6); published April 2026<sup>1</sup>

**Self-report runs hot; neutral measurement runs cold; both are true.** 80 percent of technical leaders say agents already deliver measurable financial impact and 57 percent run multi-step workflows (Anthropic and Material, n=500+, late 2025)<sup>6</sup>; neutral staging finds scaled agent use in single digits per function (AI Index 2026)<sup>1</sup>. Presence is broad, instrumented scale is rare. Ask which one your dashboard measures.



# What the frontier can do, in eight tests

**Benchmarks are the industry's standardized exams, and the fastest early warning you have.** Each is a fixed set of tasks with objective grading, so models can be compared across labs and across time. They measure the ceiling under ideal conditions rather than your floor in production, but when a benchmark your work depends on jumps, as software engineering did with an 18-point leap in six months, your target levels are due for review.

**Exhibit 9**

**Released frontier models now clear most human baselines; the open gaps are exactly where work is long and messy**

Best published score by a released frontier model, % of tasks solved, August 2026 unless noted; dashed marks: human baselines where measured; violet bars: scores refreshed on live leaderboards in August 2026

**KNOWLEDGE AND REASONING**

**Humanity's Last Exam**

2,500 questions written by domain experts to be the hardest gradeable academic exam. Text-only score; tool-augmented runs land higher.

53.3 · Claude Fable 5

**GPQA Diamond**

PhD-level science questions; domain PhDs score ~65 even with the web. Effectively saturated.

94.3 · Gemini 3.1 Pro

domain PhDs ~65

**ARC-AGI-2**

Novel abstract grid puzzles, easy for people and built to resist memorization; the standing test of fluid reasoning.

87.9 · Claude Opus 4.8, May 2026

avg tester ~60

**AIME**

US olympiad-qualifier mathematics. Saturated in December 2025 and retired as a differentiator; benchmarks age fast.

100 · GPT-5.2 Pro, Dec 2025

**SOFTWARE ENGINEERING**

**SWE-bench Verified**

Fix real reported bugs in real open-source repositories, graded by each project's own test suite. Was 76.8 in February 2026.

95.0 · Claude Fable 5

**Terminal-Bench 2.0**

End-to-end jobs in a live terminal: builds, scripts, system administration. The shared reference in frontier model cards.

77.3 · frontier best, Apr 2026

**ACTING IN THE WORLD**

**OSWorld-Verified**

Full desktop computer use across real applications. Top five models sit within 1.1 points; humans solve ~72 of the original suite.

86.1 · Qwen3.8 Max, Aug 2026

**GAIA**

Everyday assistant requests that need browsing, tools, and several steps; conceptually simple for people, humans score 92.

74.5 · frontier best, Apr 2026

human 92

Scores are the best published result by a released model on each benchmark's canonical split under the leaderboard's scaffold; unreleased preview models lead several boards and are excluded here. Blind human preference agrees the race is tight: the top four labs sit within 25 Elo points (LMArena, March 2026)<sup>31</sup>. For balance, keep the jagged edge in view: on reading analog clocks, frontier models score 50.1 against a human 90.1 (ClockBench, AI Index 2026)<sup>1</sup>.

Source: LLM-Stats and BenchLM leaderboard aggregators, August 3-5, 2026<sup>32</sup>; OSWorld-Verified leaderboard<sup>33</sup>; Stanford HAI AI Index 2026 (April) for Terminal-Bench and GAIA<sup>1</sup>; benchmark definitions per the original papers

**How to monitor without drowning: four habits.** Track one preference board (LMArena), one composite index (Artificial Analysis or LLM-Stats), and the one task suite closest to your work, at a quarterly cadence. Distrust any score quoted without a date and a scaffold, since both move results by double digits. Retire a benchmark from your dashboard once the frontier clusters within a point or two, the way competition math did. Above all, build a private eval from your own workflows: public benchmarks measure the frontier's ceiling, and only your traces measure your floor.

# Nine layers, one honest picture of where you stand

**Place your organization row by row, with an artifact as evidence for every placement.** The registry, the cost report, the eval run: no artifact, no level. Your level is the median of the nine rows; your risk is any row two or more levels below it; your plan is the lagging rows' upgrade moves, taken in dependency order. Identity (5) and observability (7) come first in every workable sequence, and evals (8) set how fast autonomy can safely widen.

## Exhibit 10

### Identity and cost observability come before everything above Level 4; evals set the speed of autonomy

Observable state per layer and level<sup>1</sup>

|                                                                                                                                              | Level 0-1<br>Shadow<br><i>Unmanaged, unofficial</i> | Level 2<br>Sanctioned<br><i>Human works, AI drafts</i> | Level 3<br>Systematic<br><i>Human drives, AI assists</i> | Level 4<br>Instrumented<br><i>AI executes, human approves</i> | Level 5<br>Orchestrated<br><i>AI executes, human on exceptions</i> | Level 6<br>Agentic-first<br><i>AI operates, human governs</i> |
|----------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|--------------------------------------------------------|----------------------------------------------------------|---------------------------------------------------------------|--------------------------------------------------------------------|---------------------------------------------------------------|
|  <b>1 Models &amp; inference</b><br>Configuration           | Personal accounts only                              | 1-2 vendor suites, default models                      | 3+ vendors, keys per team, choice by habit               | Gateway portfolio, routing, virtual keys                      | Self-hosted specialists, eval-driven routing                       | Task-tuned fleet, continuous retraining                       |
|  <b>2 Orchestration &amp; architecture</b><br>Configuration | Copy-paste between apps                             | Single-turn prompting in vendor UIs                    | First chains and no-code flows, brittle                  | Deterministic pipelines, gates built in                       | Orchestrator-worker patterns, supervisors                          | Agent meshes, agent-first process design                      |
|  <b>3 Knowledge, context &amp; memory</b><br>Context        | Knowledge in heads, pasted ad hoc                   | Chat-with-docs on scattered sources                    | Per-team RAG, duplicated effort                          | Governed, permission-aware sources, reused                    | Company brain, managed agent memory                                | Self-maintaining knowledge substrate                          |
|  <b>4 Tools &amp; actions</b><br>Control, Configuration    | None, AI only talks                                 | AI reads and drafts, humans act                        | First function calls, per-app bots                       | Tool registry, MCP connectors, writes gated                   | Cross-system action, escalation not approval                       | Agent-to-agent transactions                                   |
|  <b>5 Identity, permissions &amp; security</b><br>Control | Data leaves via personal prompts                    | SSO and DLP for humans, AI has no identity             | Shared keys, borrowed OAuth, no delegation chain         | Per-agent identity, kill switch, audit                        | Delegation chains, sandbox by default                              | Identity fabric incl. external agents                         |
|  <b>6 Deployment &amp; operations</b><br>Configuration    | SaaS defaults                                       | Vendor cloud only                                      | Laptop scripts, pilots run where built                   | Sandboxes, staging, checkpoints                               | Canaries, failover, agents watching agents                         | Autonomous operations within policy                           |
|  <b>7 Observability &amp; cost</b><br>Cost                | Spend hidden in expenses                            | License invoices, seat counts                          | Token dashboards without attribution                     | Full traces, cost per agent, team, use case                   | Cost per outcome drives routing                                    | Real-time AI P&L                                              |
|  <b>8 Evals &amp; improvement</b><br>Cultivation          | Vibes                                               | Thumbs up or down                                      | Manual spot checks, "pilot went well"                    | Golden sets, gated changes, calibrated judges, replay         | Online evals, loop closed to tuning                                | Self-improving fleet under eval governance                    |
|  <b>9 People &amp; operating model</b><br>Cultivation     | Enthusiasts self-teach                              | IT admin owns licenses, generic training               | AI lead and CoE, skills pool with power users            | Named owner per workflow and agent, trace-based coaching      | Orchestration as a role, fleet managers                            | Organization designed around fleets                           |

<sup>1</sup>Column shading deepens with the adoption level; icons mark what each stack layer covers. Three themes run across all rows rather than owning one: data quality, governance and compliance, and use-case value selection.

Source: Agentic Adoption Levels (Levels 0-6) and Agentic Stack Matrix, Hildesheim and Hainzl, August 2026; design conventions per CMMI capability profiles, SAE J3016 responsibility levels, NIST CSF profiles, CISA Zero Trust Maturity Model<sup>34</sup>

**1. Place, with evidence.** Per row, pick the cell that matches reality, and name the artifact that proves it.

**2. Read the profile.** Median = your level. Spread = your risk. The lagging rows, not the average, set the agenda.

**3. Move in dependency order.** Identity (5) and observability (7) first, evals (8) second, autonomy (4, 6) after, models (1) and people (9) continuously. If you take one action from this page: bring rows 5 and 7 to Level 4 before scaling anything else.

**The mental model behind this matrix.** Determine an agentification level for every serious value-creation flow in the organization. Many processes should not exist at all going forward; agents do not belong everywhere. And adoption does not happen at the level of the organization or the tool: it happens at the level of the workflow.

# Models and inference

L CO-AUTHOR **To be announced** · Founder, Lyceum



Lyceum is a European AI cloud from Berlin: serverless inference on open models plus dedicated GPU capacity, with EU data residency. The practitioner view of the inference stack follows in the next revision.

37%

of enterprises already run five or more foundation models in production (a16z CIO survey, June 2025)

1/68

unit cost of our task-trained 9B open model vs the strongest frontier configuration on the same workflow, at higher quality (fermisense.com, July 2026)

**Multi-model is now the enterprise default, which makes routing the real decision.**

Enterprises deploy three or more foundation models and match them to use cases by performance and cost. Vendor share shifts fast: Anthropic overtook OpenAI in enterprise API usage during 2025, and per-token prices keep falling while consumption rises faster. Any architecture hard-wired to one provider is a liability; the question changes from "which model" to "who decides which model, per task, on what evidence".

**Specialist models turn inference from a cost line into an asset.** Frontier APIs win on general capability, but task-trained small models beat them on narrow, high-volume workflows at a fraction of the cost, and they run on hardware you own. The route there runs through the gateway: once traffic is routed and measured, the high-volume tasks that justify a specialist identify themselves.

**Exhibit 11**

**The model landscape splits into four segments; the gateway segment is where control begins**

Selected players per segment, not exhaustive<sup>1</sup>

|                                                                                                                       |            |             |                 |      |               |
|-----------------------------------------------------------------------------------------------------------------------|------------|-------------|-----------------|------|---------------|
| <b>Frontier model APIs</b><br>Maximum capability, fastest start; premium pricing and a data boundary to manage        | OpenAI     | Anthropic   | Google DeepMind | xAI  | Cohere        |
| <b>Open-weights families</b><br>Control, residency, and unit cost at volume; you own quality management               | Meta Llama | Mistral     | DeepSeek        | Qwen | Moonshot Kimi |
| <b>Gateways and routing</b><br>One neutral chokepoint: keys, budgets, allowlists, fallbacks, task-based routing       | LiteLLM    | OpenRouter  | Portkey         | Kong | Cloudflare    |
| <b>Serving and tuning</b><br>Self-hosted inference and the specialist-model loop, from serving engines to fine-tuning | vLLM       | Together AI | Fireworks       | Groq | Predibase     |
|                                                                                                                       | Cerebras   | Baseten     |                 |      |               |

<sup>1</sup>Logos are the property of their respective owners and appear for identification only.

Source: Menlo Ventures, State of Generative AI in the Enterprise (November 2024, December 2025)<sup>8</sup>; a16z enterprise CIO survey (June 2025)<sup>15</sup>

| ARCHETYPE                         | VALUE PROPOSITION                                                                  | CHOOSE WHEN                                                  |
|-----------------------------------|------------------------------------------------------------------------------------|--------------------------------------------------------------|
| <b>Frontier API first</b>         | Best general capability, zero infrastructure, newest features earliest             | Low volume, exploratory work, frontier reasoning             |
| <b>Gateway-mediated portfolio</b> | Neutral control point: swap models without code change, budgets and policy per key | From Level 4 onward, always; prerequisite for layers 5 and 7 |
| <b>Self-hosted open weights</b>   | Data residency, predictable unit cost, no external dependency                      | Sensitive data, high volume, sovereignty needs               |
| <b>Task-tuned specialists</b>     | Beats frontier per dollar on narrow workflows, runs on owned hardware              | Stable, high-volume tasks with measurable quality            |

**WHAT GOOD LOOKS LIKE, BY LEVEL**

|                                            |                                                 |                                                  |                                                  |                                               |
|--------------------------------------------|-------------------------------------------------|--------------------------------------------------|--------------------------------------------------|-----------------------------------------------|
| <b>Level 1</b><br>Vendor suites as shipped | <b>Level 2</b><br>Team keys, habit-based choice | <b>Level 3</b><br>Gateway, virtual keys, routing | <b>Level 4</b><br>Specialists join the portfolio | <b>Level 5</b><br>Fleet retrained from traces |
|--------------------------------------------|-------------------------------------------------|--------------------------------------------------|--------------------------------------------------|-----------------------------------------------|

**DIAGNOSTIC** Could you swap the model behind your flagship workflow without touching code? Do you know cost and quality per model per task, or only per vendor invoice?

**SO WHAT**

Put one gateway in front of all model traffic and move the raw keys into it. Every other capability in this report attaches to that chokepoint.

# Intelligence ownership: task-trained specialists clear the frontier plateau

## 87.3%

of the maximum achievable score reached by the authors' GRPO-trained 9B open model on catalog review; five frontier models plateaued at 69.7 to 76.9 percent (fermisense.com, July 2026)

## 2.2x

revenue growth of the top quartile of AI spenders from November 2022 to December 2025, against roughly 15 percent for firms with zero AI spend (Ramp data, via fermisense.com)

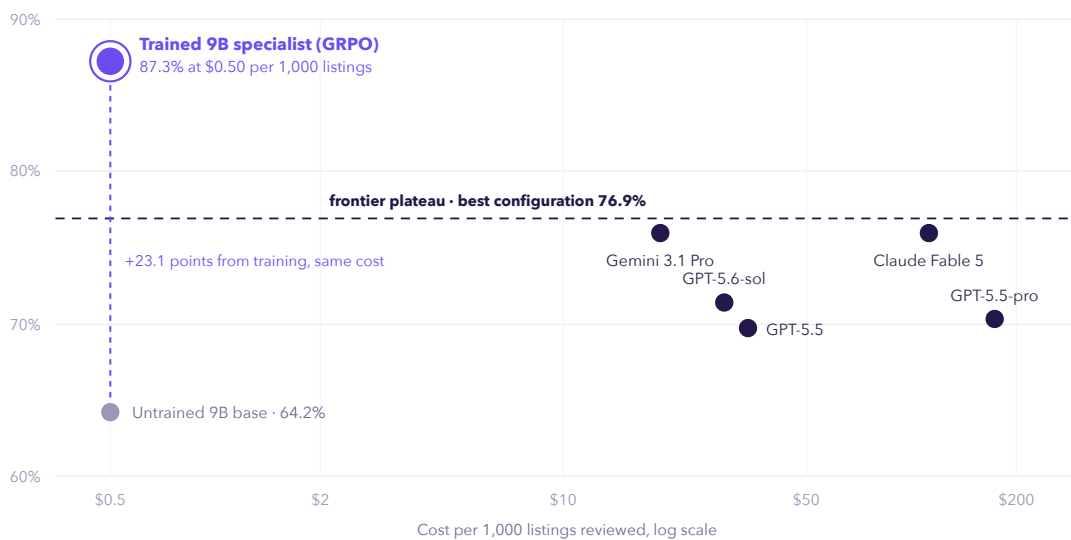
**The winners converge on one playbook: an open model, proprietary task data, and reinforcement learning against a scored copy of the workflow.** Bridgewater reports a trained model making roughly 30 percent fewer mistakes than the best frontier model, Harvey's legal agent beats frontier models on its own rubrics, and Intercom's Fin resolves more support issues at lower cost. Most organizations get there pragmatically: 47 percent combine off-the-shelf agents with custom-built components rather than choosing one or the other (Anthropic and Material, late 2025).

**Our own published experiment shows the economics.** On e-commerce catalog integrity with 200 held-out episodes, a 9B open model trained with GRPO for about \$500 (two workstation GPUs, 1,000 steps, three and a half days) cleared every frontier configuration we tested by more than 10 points, at \$0.50 per 1,000 listings: 40 times cheaper than the least expensive frontier setup, 68 times cheaper than the strongest, and roughly 340 times cheaper than the most expensive. The five frontier models plateaued within a tenth of a point of each other; training on the task cleared that ceiling.

### Exhibit 12

## On one instrumented workflow, the trained specialist beats every frontier configuration at a fraction of the cost

Share of maximum achievable score vs cost per 1,000 listings, e-commerce catalog integrity, 200 held-out episodes<sup>1</sup>



<sup>1</sup>Same tools, images, and scorer for every configuration; frontier models run with optimized prompts. Specialist: Qwen3.5-9B trained with GRPO via prime-rl. Source: Hildesheim, Hainzl et al., "The Rise of Intelligence Ownership", fermisense.com, July 27, 2026<sup>21</sup>

### SO WHAT

Once a workflow is instrumented and evaluated at Level 4, its traces become training data. Specialists are how Level 5 economics arrive: quality above the frontier plateau at commodity cost, on hardware you own.

# Orchestration and architecture



CO-AUTHOR **Jakob Mayer** · Founder and CEO, OSM Data



OSM Data

**OSM Data** builds mathematical optimization systems for energy planning, trading, scheduling, and real-time operations, connecting constrained decisions across markets and time horizons. Deterministic decision systems of exactly this kind are the rails this layer wraps around agent workflows.

16%

of enterprise "agent" deployments qualify as true agents; most are fixed-sequence workflows (Menlo Ventures, December 2025)

~90%

of transformative, function-specific use cases remain stuck in pilot (McKinsey, June 2025)

**The framework wars matter less than the control-flow decision.** Whether control flow lives in human clicks, in code, or in the model is the real architectural choice at every level. The production doctrine across credible practice is consistent: build the simplest thing that works, use workflow patterns before autonomous agents, and add complexity only when a measured gap demands it.

**Most production "agents" are workflows, and that is fine.** The failure mode is not too little autonomy but the god-agent that owns everything and fails like a company where every decision routes through one person. Bounded ownership per agent, on a deterministic spine, with AI reserved for genuine judgment points, is the architecture that survives contact with audit, scale, and cost.

## Exhibit 13

### Four segments serve different builders; mature stacks combine a code-first core with deterministic rails

Selected players per segment, not exhaustive<sup>1</sup>

#### Code-first agent frameworks

Full control and testability for engineering teams; the de facto standard layer

LangGraph LlamaIndex CrewAI AutoGen, Semantic Kernel  
 Pydantic AI Haystack Google ADK Mastra

#### Agent runtimes

Packaged agentic loops for knowledge and code work; fastest capability per hour invested

Claude Code, Cowork OpenAI Agents SDK Copilot Studio

#### Deterministic rails

Durable execution and integration spine that agent workflows run on

Temporal Airflow n8n Zapier Inngest

#### Suite agent platforms

Agents embedded in systems you already run; speed inside, lock-in at the boundary

Agentforce ServiceNow UiPath Microsoft 365 SAP Joule

IBM Watsonx

<sup>1</sup>Logos are the property of their respective owners and appear for identification only.

Source: Anthropic, Building Effective Agents (December 2024)<sup>22</sup>; OpenAI, A Practical Guide to Building Agents (2025)<sup>23</sup>; Menlo Ventures (December 2025)<sup>8</sup>

| ARCHETYPE                                         | VALUE PROPOSITION                                           | CHOOSE WHEN                                                   |
|---------------------------------------------------|-------------------------------------------------------------|---------------------------------------------------------------|
| <b>Deterministic spine, AI at judgment points</b> | Auditable, testable, cheap; AI only where rules run out     | Regulated and high-stakes workflows; the default from Level 4 |
| <b>Code-first framework</b>                       | Control over state, retries, gates; portable across vendors | You have engineers and the workflow is core IP                |
| <b>No-code flows</b>                              | Speed and citizen builders; good for discovery              | Level 3 experimentation; expect a rebuild before production   |
| <b>Suite agents</b>                               | Native data access, fast time to value inside one system    | The workflow lives entirely inside that suite                 |

#### WHAT GOOD LOOKS LIKE, BY LEVEL

| Level 1                     | Level 2                        | Level 3                           | Level 4                      | Level 5                          |
|-----------------------------|--------------------------------|-----------------------------------|------------------------------|----------------------------------|
| Prompting inside vendor UIs | First chains and no-code flows | Instrumented pipelines with gates | Orchestrator-worker patterns | Agent meshes, agent-first design |

**DIAGNOSTIC** Can the flagship workflow run end to end without a person ferrying context between tools?

#### SO WHAT

Rebuild the one or two workflows that matter as instrumented pipelines with explicit human gates; kill the rest of the pilot zoo.

# Knowledge, context, and memory



CO-AUTHOR **Justinas Zaliaduonis** · Founder, fermisense



**fermisense** is the authors' applied-research lab: task-trained specialist models from instrumented workflow traces. The intelligence-ownership experiment on page 12 is its work.

51%

of enterprises had adopted retrieval-augmented generation by late 2024, up from 31% a year earlier (Menlo Ventures, November 2024)

90%+

of companies have workers regularly using personal AI tools, feeding company context into ungoverned channels (MIT NANDA, 2025)

**The maturity path is retrieval sophistication plus governance, not more indexes.** The field progressed from naive retrieval-augmented generation through advanced and modular retrieval to graph-based and agentic retrieval, where the model decides when and how to retrieve. The enterprise problem differs from the demo problem: permission-aware retrieval, provenance per answer, and one governed source reused across workflows instead of five bespoke builds.

**Context engineering has replaced prompt engineering as the discipline.** The practice: curate the smallest set of high-signal tokens, compact aggressively, keep notes outside the window, retrieve just in time. Agent memory is becoming a managed service with retention rules, and memory poisoning is now a named attack class to defend against.

## Exhibit 14

### Four segments, one test: does company knowledge reach the model governed, permissioned, and reused

Selected players per segment, not exhaustive<sup>1</sup>

|                                                                                                                          |          |          |                  |          |             |         |
|--------------------------------------------------------------------------------------------------------------------------|----------|----------|------------------|----------|-------------|---------|
| <b>Vector and hybrid stores</b><br>The retrieval substrate; differences are operating model and hybrid-search depth      | Pinecone | Weaviate | Qdrant           | pgvector | Redis       | MongoDB |
|                                                                                                                          | Chroma   | Milvus   | LanceDB          |          |             |         |
| <b>Enterprise index and search</b><br>Permission-aware, cross-tool retrieval as a product                                | Glean    | Elastic  | Microsoft Search | Coveo    | Onyx        |         |
| <b>Graph and structured context</b><br>When relations carry the meaning: entities, hierarchies, cross-document reasoning | Neo4j    | GraphRAG | LlamaIndex       |          |             |         |
| <b>Agent memory layers</b><br>Persistent episodic and semantic memory across sessions, with temporal validity            | Letta    | Zep      | Mem0             | Cognee   | Supernemory |         |

<sup>1</sup>Logos are the property of their respective owners and appear for identification only.

Source: Gao et al., Retrieval-Augmented Generation survey (2023)<sup>19</sup>; Microsoft GraphRAG (2024)<sup>24</sup>; Anthropic, Effective Context Engineering (September 2025)<sup>22</sup>; Zep (2025)<sup>24</sup>

| ARCHETYPE                       | VALUE PROPOSITION                                              | CHOOSE WHEN                                             |
|---------------------------------|----------------------------------------------------------------|---------------------------------------------------------|
| <b>DIY retrieval per team</b>   | Full flexibility, fastest first demo                           | Level 3 exploration; consolidate before it multiplies   |
| <b>Managed enterprise index</b> | Permission-aware retrieval across tools, bought not built      | Broad access matters more than custom pipelines         |
| <b>Governed context sources</b> | One connector per system of record, reused by every workflow   | From Level 4; the reuse pattern that compounds          |
| <b>Memory service</b>           | Agents that remember across sessions, with audit and retention | Level 5 long-running agents; govern like any data store |

#### WHAT GOOD LOOKS LIKE, BY LEVEL

| Level 1                      | Level 2                         | Level 3                            | Level 4                       | Level 5                    |
|------------------------------|---------------------------------|------------------------------------|-------------------------------|----------------------------|
| Copy-paste context, app ACLs | Per-team RAG, duplicated effort | Governed, permission-aware sources | Company brain, managed memory | Self-maintaining substrate |

**DIAGNOSTIC** *Is retrieval permission-aware, or can the index see more than the asking user should?*

#### SO WHAT

Stand up one governed, permission-aware context source and make two workflows share it.

# Tools and actions



CO-AUTHOR **Open slot** · practitioner co-authors welcome: [fabian.hildesheim@cdtm.com](mailto:fabian.hildesheim@cdtm.com)

A founder building in this layer adds their photo, a short company introduction in this space, and one review pass of the page.

40%

of enterprise applications will ship task-specific agents by end-2026, up from under 5% in 2025 (Gartner, August 2025)

1M+

custom agents created by Microsoft customers in a single quarter, most outside IT review (Microsoft Build, May 2025)

**The protocol layer has consolidated, which de-risks the build.** MCP, the Model Context Protocol for connecting agents to tools, went from launch in late 2024 to an OAuth-grade security specification, a public registry with private enterprise sub-registries, and neutral governance under the Linux Foundation, alongside the A2A protocol for inter-agent traffic. Betting on MCP is now the low-risk choice; the enterprise pattern is an allowlisted private sub-registry with scoped tokens per tool.

**Design tools like an attack surface, because they are one.** Few, sharp, well-described tools beat many vague ones: every tool is decision noise for the model and reach for an attacker. Reads can run broad; writes are gated by action class until evals earn escalation-only supervision. Computer use covers the legacy systems without APIs, at the price of sandboxing discipline.

## Exhibit 15

### Standards now anchor the action layer; platforms and computer use extend its reach

Selected players per segment, not exhaustive<sup>1</sup>

#### Protocols and registries

The standards spine: connectivity, discovery, and inter-agent communication

MCP and MCP Registry A2A, Agentic AI Foundation OpenAPI Smithery

#### Integration platforms

Thousands of prebuilt actions; fastest surface, needs registry governance

Zapier Make n8n Workato Composio Arcade Pipedream

#### Computer use and browsing

Agents operating user interfaces directly; reach into systems that will never get an API

Anthropic OpenAI Operator Browser Use Browserbase

#### RPA bridge

The existing bot estate: deterministic, brittle, increasingly orchestrated by agents

UiPath Automation Anywhere Power Automate

<sup>1</sup>Logos are the property of their respective owners and appear for identification only.

Source: Model Context Protocol specification and registry (2024-2025)<sup>25</sup>; Linux Foundation, Agentic AI Foundation and A2A (2025)<sup>26</sup>; Gartner (August 2025)<sup>14</sup>

| ARCHETYPE                           | VALUE PROPOSITION                                                | CHOOSE WHEN                                           |
|-------------------------------------|------------------------------------------------------------------|-------------------------------------------------------|
| <b>MCP-first connectors</b>         | Standardized, governable, reusable across every agent and vendor | Core systems of record; the Level 4 default           |
| <b>Integration-platform actions</b> | Breadth today: thousands of SaaS actions without build effort    | Long tail of tools, wrapped under registry governance |
| <b>Computer use</b>                 | Universal reach, including legacy user interfaces                | No API exists and volume justifies the latency        |
| <b>RPA under agent control</b>      | Reuses the bot estate you already paid for                       | Stable deterministic subtasks inside agent workflows  |

#### WHAT GOOD LOOKS LIKE, BY LEVEL

| Level 1              | Level 2                            | Level 3                     | Level 4                         | Level 5                     |
|----------------------|------------------------------------|-----------------------------|---------------------------------|-----------------------------|
| AI reads, humans act | First function calls, per-app bots | Registry, MCP, writes gated | Cross-system action, escalation | Agent-to-agent transactions |

**DIAGNOSTIC** Can an agent change a record in a business system today, and under whose credentials? Could you list every tool every agent can call?

#### SO WHAT

Stand up the tool registry and per-agent scoping before expanding what agents can touch. An action surface without identity is an incident schedule.



# Computer use and reinforcement learning



CO-AUTHOR **Ingmar Klein** · Founder, Huzzle Labs



**Huzzle Labs**

**Huzzle Labs** trains small computer-use models in proprietary reinforcement-learning environments, built with a network of 300,000+ domain experts. Its h-1 family ranks beside far larger frontier models on OSWorld and keeps a fully local, replayable audit trail.

**66.3%**

top score on OSWorld real-computer tasks, within six points of the 72 percent human baseline and up from 12 percent two years earlier (AI Index 2026)

**8B**

parameters in Huzzle Labs' h-1 computer-use model, trained in its own RL environments and ranking beside far larger frontier models on OSWorld

**Computer use turns the screen itself into the API.** Agents that see pixels and operate keyboard and mouse reach the systems that will never expose an interface: the plant terminal, the decades-old ERP screen, the supplier portal. Capability has moved from party trick to near-human on real desktop tasks in two years, which changes the reach of Levels 4 and 5: the automatable surface now includes the software your integration roadmap had written off.

**A specialist ecosystem is forming beside the frontier.** Anthropic and OpenAI ship general computer-use agents; open-source browser agents cover the web; and small task-trained models are arriving: Huzzle Labs trains its Huzzle World family (1B to 8B) in proprietary RL environments built with a network of 300,000+ domain experts, and Coast by Attention Engineering reads the screen instead of recording clicks, recovers from its own missteps mid-run, and leaves a timestamped, replayable audit trail with fully local memory. That audit-first, self-hosted profile is exactly what regulated European enterprises need from this layer.

## Exhibit 16

### The computer-use stack: frontier generalists, open-source browsers, and audit-first specialists

Selected players per segment, not exhaustive<sup>1</sup>

#### Frontier computer use

General-purpose screen agents from the labs; broad capability, cloud-hosted

Anthropic computer use OpenAI Operator Gemini Computer Use  
 Amazon Nova Act UI-TARS

#### Open-source browser agents

Web-surface automation owned in your repository

Browser Use E2B sandboxes Skyvern Steel

#### Specialist challengers

Small task-trained models and audit-first agents, self-host friendly

Huzzle Labs · h-1 Coast · Attention Engineering Manus

#### RPA estate under agent control

The deterministic bot fleet, increasingly orchestrated rather than replaced

UiPath Automation Anywhere

<sup>1</sup>Logos are the property of their respective owners and appear for identification only.

Source: Stanford HAI AI Index 2026 (Section 2.6)<sup>1</sup>; Huzzle Labs (labs.huzzle.com)<sup>35</sup>; Attention Engineering (attention.inc)<sup>35</sup>; vendor documentation

| ROUTE                          | VALUE PROPOSITION                                        | CHOOSE WHEN                                                            |
|--------------------------------|----------------------------------------------------------|------------------------------------------------------------------------|
| <b>API or MCP first</b>        | Deterministic, fast, cheap, fully auditable              | An interface exists or can be built; always the default                |
| <b>Computer use</b>            | Universal reach, including legacy user interfaces        | No API will ever exist and volume justifies latency; sandbox mandatory |
| <b>RPA under agent control</b> | Reuses the deterministic bot estate you already paid for | Stable, unchanging screens inside agent workflows                      |

#### SO WHAT

Treat computer use as reach, not default: APIs first, screen agents for the systems that will never get one, always inside a sandbox with gated writes and a replayable trail.



# Identity, permissions, and security

K

CO-AUTHOR **To be announced** · Founder, [kontext.security](#)

 **kontext.security**

**kontext.security** co-authors this layer. Company introduction and the practitioner view of agent identity and containment follow in the next revision.

**82%**  
of enterprises have discovered AI agents that IT did not know existed (Cloud Security Alliance, 2026)

**109:1**  
machine identities per human identity in the enterprise; roughly 79 of the 109 are AI agents (Palo Alto Networks, 2026)

**The field data says this layer decides production success.** 68 percent of organizations cannot distinguish agent actions from human actions, and 35 percent admit they could not shut down a rogue agent. The 2025-2026 incident record, from stolen OAuth tokens to over-permissioned coding agents deleting production databases to zero-click copilot exfiltration, is uniformly an access story, not a model story. Government guidance now requires defining who owns, approves, monitors, and can stop each agent before deployment.

**Three security philosophies compete, and mature programs run all three.** Identity-first gives every agent a directory identity with scoped, short-lived credentials and a delegation chain. Firewall-first inspects prompts and outputs for injection and leakage. Containment-first assumes a confusable agent and controls what it can reach. Identity is the one you cannot retrofit, which is why it gates the matrix.

Exhibit 17

Identity, guardrails, containment, and standards form the control stack for agents

Selected players per segment, not exhaustive<sup>1</sup>

|                                                                                                                      |                                                                                                                 |                                                                                                     |                                                                                                     |                                                                                             |                                                                                                       |
|----------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| <b>Agent identity and access</b><br>First-class agent identities, scoped credentials, lifecycle, quarantine          |  Entra Agent ID, Agent 365     |  Okta              |  Auth0 for AI      |  WorkOS  |  Astrix            |
|                                                                                                                      |  Token Security              |  Aembit          |                                                                                                     |                                                                                             |                                                                                                       |
| <b>Guardrails and AI firewalls</b><br>Inspect and filter model inputs and outputs: injection, leakage, policy checks |  Lakera                      |  Protect AI      |  NeMo Guardrails |  Lasso |  Prompt Security |
|                                                                                                                      |  HiddenLayer                 |  Guardrails AI   |                                                                                                     |                                                                                             |                                                                                                       |
| <b>Containment and secrets</b><br>Sandboxes, secret management, rotation; control reach, not intentions              |  E2B                         |  HashiCorp Vault |  sandbox-runtime |                                                                                             |                                                                                                       |
| <b>Standards and guidance</b><br>The checklists and certifiable anchors auditors will ask about                      |  OWASP GenAI, Agentic Top 10 |  NIST AI RMF     |  ISO IEC 42001 |                                                                                             |                                                                                                       |
|                                                                                                                      |  Five Eyes, NCSC             |  MITRE ATLAS     |                                                                                                     |                                                                                             |                                                                                                       |

<sup>1</sup>Logos are the property of their respective owners and appear for identification only.  
Source: Cloud Security Alliance (2026)<sup>29</sup>; Palo Alto Networks (2026)<sup>29</sup>; OWASP GenAI Security Project (2024-2025)<sup>27</sup>; Five Eyes (April 2026)<sup>28</sup>; NCSC (May 2026)<sup>28</sup>

| ARCHETYPE                | VALUE PROPOSITION                                                        | CHOOSE WHEN                                        |
|--------------------------|--------------------------------------------------------------------------|----------------------------------------------------|
| <b>Identity-first</b>    | Per-agent identity, delegation chain, kill switch; audit by construction | Always, from Level 4; cannot be retrofitted later  |
| <b>Firewall-first</b>    | Fast to deploy, catches known injection and leakage patterns             | Additive layer; never a substitute for scoping     |
| <b>Containment-first</b> | Assumes compromise, limits blast radius at the environment layer         | Any agent that executes code or touches production |
| <b>Standards-led</b>     | Certifiable governance and audit-ready checklists                        | Regulated industries, board and auditor visibility |

WHAT GOOD LOOKS LIKE, BY LEVEL

|                                               |                                               |                                                          |                                                      |                                                         |
|-----------------------------------------------|-----------------------------------------------|----------------------------------------------------------|------------------------------------------------------|---------------------------------------------------------|
| <b>Level 1</b><br>SSO and DLP for humans only | <b>Level 2</b><br>Shared keys, borrowed OAuth | <b>Level 3</b><br>Per-agent identity, kill switch, audit | <b>Level 4</b><br>Delegation chains, sandbox default | <b>Level 5</b><br>Identity fabric incl. external agents |
|-----------------------------------------------|-----------------------------------------------|----------------------------------------------------------|------------------------------------------------------|---------------------------------------------------------|

**DIAGNOSTIC** *Could you shut down a misbehaving agent in minutes, and has that ever been tested? Can your logs distinguish agent actions from human actions?*

**SO WHAT**  
Identity per agent plus gates on writes. Together with layer 7, this is the crossing that turns pilots into governable operations.

# Deployment and operations



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A founder building in this layer adds their photo, a short company introduction in this space, and one review pass of the page.

80%

of companies using agents say agents have already taken unintended actions (SailPoint and Dimensional Research, 2025)

12x

more AI projects reach production when governance is in place than without it (Databricks, 2026)

**Progressive autonomy is the converged deployment doctrine.** Government guidance, lab guidance, and platform practice land on the same ladder: read-only first, then every state-changing action behind a gate, then policy-scoped autonomy where pre-approved action classes execute and exceptions escalate, then monitored autonomy inside a policy envelope with a working kill switch. Human intervention triggers are defined upfront: failure thresholds and high-risk actions.

**Pilot-grade infrastructure quietly carrying production load is the silent killer.** The Level 3 pattern is scripts running where they were built, with no rollback and no owner at 2am. The Level 4 pattern is boring on purpose: sandboxed runtimes, staging versus production, versioned prompts and configurations, checkpointed runs that resume instead of restarting, and incident playbooks that name AI failure modes.

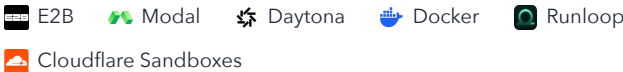
## Exhibit 18

### Sandboxes, cloud platforms, and durable rails carry agents; gate mechanics keep humans in control

Selected players per segment, not exhaustive<sup>1</sup>

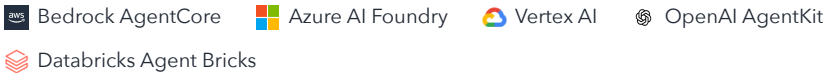
#### Sandboxed runtimes

Isolated execution for untrusted agent actions, from micro-VMs to containers



#### Cloud agent platforms

Managed agent infrastructure inside the cloud estate; IAM-native, some lock-in



#### Orchestrated infrastructure

The durable-execution and scaling substrate under the agent platform



#### Human-gate mechanics

Interrupt, approve, edit, reject, resume: checkpointed human control in the loop



<sup>1</sup>Logos are the property of their respective owners and appear for identification only.

Source: NCSC (May 2026)<sup>28</sup>; OpenAI, A Practical Guide to Building Agents (2025)<sup>23</sup>; Anthropic containment engineering (May 2026)<sup>22</sup>; SailPoint (2025)<sup>29</sup>; Databricks (2026)<sup>29</sup>

| ARCHETYPE                         | VALUE PROPOSITION                                                             | CHOOSE WHEN                                              |
|-----------------------------------|-------------------------------------------------------------------------------|----------------------------------------------------------|
| <b>Hyperscaler agent platform</b> | Integrated identity, monitoring, and scale inside the existing cloud contract | Cloud-committed estates without sovereignty constraints  |
| <b>Neutral sandbox plus rails</b> | Portable, self-host friendly, vendor-independent                              | Self-hosted control planes, mixed clouds, EU sovereignty |
| <b>DIY containers</b>             | Maximum control at maximum operating burden                                   | Strong platform team and unusual requirements            |

#### WHAT GOOD LOOKS LIKE, BY LEVEL

|                                           |                                               |                                                  |                                                    |                                                |
|-------------------------------------------|-----------------------------------------------|--------------------------------------------------|----------------------------------------------------|------------------------------------------------|
| <b>Level 1</b><br>Vendor cloud, click-ops | <b>Level 2</b><br>Laptop scripts, no rollback | <b>Level 3</b><br>Sandboxes, staging, versioning | <b>Level 4</b><br>Canaries, failover, self-healing | <b>Level 5</b><br>Autonomous ops within policy |
|-------------------------------------------|-----------------------------------------------|--------------------------------------------------|----------------------------------------------------|------------------------------------------------|

**DIAGNOSTIC** *Where does the flagship AI workflow physically run, and what happens when it fails at 2am? Could you roll back last week's prompt change?*

#### SO WHAT

Give the flagship workflow real environments, version control, and a named on-call owner. Then, and only then, widen its autonomy one rung.

# Observability and cost



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A founder building in this layer adds their photo, a short company introduction in this space, and one review pass of the page.

26%

of \$1B+ organizations have real-time visibility into what their AI agents cost (KPMG AI Pulse, June 2026)

80%

of companies miss their AI spend forecast by 25 percent or more (Ramp Economics Lab, card and bill-pay data of 70,000+ businesses, 2026)

**Nobody can manage what hides in five places.** AI spend hides in API keys, seats, cloud marketplaces, renewals with embedded agents, and expensed subscriptions. In our own interviews, zero of thirteen CIOs and CTOs with multi-million model bills could break spend down by use case. Token dashboards are not the answer: the unit of account is cost per completed task, joined to the workflow outcome in the systems where work is recorded.

**The standards and the discipline both exist now; the join is the work.** OpenTelemetry's GenAI conventions cover model, agent, and MCP telemetry, the observability tooling market is mature, and FinOps for AI defines the progression from prototyping to embedded to continuously optimized cost. What no tool ships out of the box is the attribution join from token to team to use case to outcome. That join is the system of record, and no other build in this report pays back faster.

## Exhibit 19

### Tracing is commoditizing; the attribution join from spend to outcome is the differentiating work

Selected players per segment, not exhaustive<sup>1</sup>

|                                                                                                                 |                           |               |                    |           |            |
|-----------------------------------------------------------------------------------------------------------------|---------------------------|---------------|--------------------|-----------|------------|
| <b>LLM observability, open source</b><br>Self-hostable tracing and analytics; data stays on your infrastructure | Langfuse                  | Arize Phoenix | Helicone           | Traceloop | Comet Opik |
|                                                                                                                 | AgentOps                  |               |                    |           |            |
| <b>LLM observability, SaaS</b><br>Depth and velocity as a service; evals and monitoring in one place            | LangSmith                 | Braintrust    | W&B Weave          | Galileo   | WhyLabs    |
|                                                                                                                 | Arize AX                  |               |                    |           |            |
| <b>APM-integrated</b><br>LLM signals inside the monitoring stack you already operate                            | Datadog LLM Observability | Elastic       | New Relic          | Dynatrace |            |
| <b>Standards and AI FinOps</b><br>The telemetry standard and the cost discipline, through to cost per outcome   | OpenTelemetry GenAI       | FinOps for AI | Ramp Economics Lab | Vantage   | CloudZero  |

<sup>1</sup>Logos are the property of their respective owners and appear for identification only. OpenTelemetry GenAI conventions are pre-stable; pin versions.  
Source: KPMG AI Quarterly Pulse (June 2026)<sup>13</sup>; Ramp Economics Lab (2026)<sup>15</sup>; FinOps Foundation (2026)<sup>35</sup>; Hildesheim and Hainzl validation interviews (2026)

| ARCHETYPE                              | VALUE PROPOSITION                                                | CHOOSE WHEN                                    |
|----------------------------------------|------------------------------------------------------------------|------------------------------------------------|
| <b>Open-source self-hosted tracing</b> | Full-payload traces without data leaving your boundary           | Self-hosted control planes, sensitive payloads |
| <b>SaaS observability plus evals</b>   | Fastest path to dashboards, datasets, and eval loops             | Cloud-friendly estates prioritizing speed      |
| <b>APM-integrated</b>                  | One pane with infrastructure monitoring and alerting             | An APM stack is already deployed               |
| <b>FinOps for AI practice</b>          | Cost allocation, budgets, unit economics as an operating cadence | From Level 4, run jointly with finance         |

#### WHAT GOOD LOOKS LIKE, BY LEVEL

| Level 1                  | Level 2                          | Level 3                            | Level 4                         | Level 5          |
|--------------------------|----------------------------------|------------------------------------|---------------------------------|------------------|
| Invoices and seat counts | Token dashboards, no attribution | Traces, cost per use case, budgets | Cost per outcome drives routing | Real-time AI P&L |

**DIAGNOSTIC** Could you produce last month's AI spend per team within an hour? What is the cost per completed task for the workflow that matters most?

#### SO WHAT

One trace log through the gateway, then the attribution join: token to team to use case to outcome. Everything else in this report compounds on this.

# Evals and improvement



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A founder building in this layer adds their photo, a short company introduction in this space, and one review pass of the page.

10-15%

of nearly 900 GenAI use cases drove about 80 percent of the value at Johnson & Johnson; the rest were shut down (Wall Street Journal, April 2025)

+39%

accuracy gain from reinforcement fine-tuning on graded traces in a verified enterprise tax-analysis case (OpenAI RFT, May 2025)

**The consensus ladder is simple, and most companies are on rung one.** Vibe checks, then a golden set of 50 to 200 real cases with binary pass-fail judgments and expert critiques, then evals wired into the change pipeline so no prompt or model change ships without the suite passing, then online evals on sampled production traffic with drift alarms. Twenty labeled cases from your own workflow beat any public benchmark as a start.

**Close the loop and this layer becomes an economic engine.** Production traces become datasets, datasets become fine-tuning runs, and reinforcement fine-tuning on graded traces is now a product capability with verified enterprise gains. The endpoint is the specialist-model result from layer 1: the improvement loop run to its economic conclusion. Judge with care: position bias, verbosity bias, and self-preference are documented, so calibrate model judges against human labels and never judge a model with itself.

## Exhibit 20

### Platforms, harnesses, and tuning loops now cover the full path from spot checks to self-improvement

Selected players per segment, not exhaustive<sup>1</sup>

|                                                                                                                |                       |                  |                             |          |                   |
|----------------------------------------------------------------------------------------------------------------|-----------------------|------------------|-----------------------------|----------|-------------------|
| <b>Eval platforms</b><br>Datasets, judges, experiments, and regression gates as a product                      | Braintrust            | LangSmith        | Langfuse                    | Galileo  | DeepEval          |
|                                                                                                                | Patronus AI           |                  |                             |          |                   |
| <b>Open-source harnesses</b><br>Pipeline-friendly test runners and retrieval metrics, owned in your repository | promptfoo             | Ragas            | W&B Weave                   | MLflow   | Inspect · UK AISI |
| <b>Tuning and RL loops</b><br>From graded traces to better and cheaper models                                  | OpenAI RFT            | Predibase        | Together AI                 | OpenPipe | Tinker            |
| <b>Practice canon</b><br>Look at your data, binary judgments, calibrated judges, trajectory grading            | Husain evals doctrine | Yan eval process | Anthropic agent-evals guide |          |                   |

<sup>1</sup>Logos are the property of their respective owners and appear for identification only.

Source: Husain (2024)<sup>24</sup>; Yan (2024-2025)<sup>24</sup>; Zheng et al. (2023)<sup>20</sup>; Anthropic, Demystifying Evals for AI Agents (January 2026)<sup>22</sup>; OpenAI reinforcement fine-tuning (May 2025)<sup>23</sup>

| ARCHETYPE                            | VALUE PROPOSITION                                             | CHOOSE WHEN                                            |
|--------------------------------------|---------------------------------------------------------------|--------------------------------------------------------|
| <b>Golden set plus gated changes</b> | Silent regressions become impossible; changes ship with proof | The Level 4 entry ticket for every flagship workflow   |
| <b>Online evals</b>                  | Sampled production scoring, drift and failure taxonomy        | Level 5, once traffic volume justifies it              |
| <b>Judge calibration</b>             | Model judges you can actually trust, anchored to human labels | Whenever a model grades a model                        |
| <b>RL on traces</b>                  | Compounding quality and falling unit cost on owned tasks      | Stable high-volume workflows with clear reward signals |

#### WHAT GOOD LOOKS LIKE, BY LEVEL

|                                     |                                             |                                                      |                                             |                                                     |
|-------------------------------------|---------------------------------------------|------------------------------------------------------|---------------------------------------------|-----------------------------------------------------|
| <b>Level 1</b><br>Thumbs up or down | <b>Level 2</b><br>Spot checks and anecdotes | <b>Level 3</b><br>Golden sets, gated changes, replay | <b>Level 4</b><br>Online evals, closed loop | <b>Level 5</b><br>Self-improvement under governance |
|-------------------------------------|---------------------------------------------|------------------------------------------------------|---------------------------------------------|-----------------------------------------------------|

**DIAGNOSTIC** *If someone changed a prompt today, what would break silently, and would you know? Has anyone seeded known errors to test whether reviewers catch them?*

#### SO WHAT

Build a golden set from 20 real cases of the flagship workflow and gate every change on it. This layer is the throttle that safely opens layers 4 and 6.

# People and operating model



CO-AUTHOR **Open slot** · practitioner co-authors welcome: [fabian.hildesheim@cdtm.com](mailto:fabian.hildesheim@cdtm.com)

A founder building in this layer adds their photo, a short company introduction in this space, and one review pass of the page.

3x

employees use AI at three times the level their leadership believes (McKinsey Superagency, January 2025)

13%

of employees see agents deeply integrated in daily workflows; frontline adoption has plateaued (BCG AI at Work, June 2025)

**The people layer is where rollouts stall, measurably.** Leaders underestimate employee AI usage roughly threefold, frontline regular use plateaus around half the workforce, and high performers differ on one behavior more than any other: they fundamentally redesigned workflows rather than adding AI beside them. Rollout is not adoption, and training disconnected from real work does not transfer.

**The operating model converges on named ownership plus platform-supported federation.** Centrally led models dominate today; the mature shape is a hub that owns the control plane, standards, and evals, with federated delivery in the business. Every agent gets a named human owner, now both government guidance and product default, and orchestration becomes a recognized role: small teams supervising agent fleets, with the human-agent ratio as a management metric.

## Exhibit 21

### This layer is patterns and roles more than products; the thin vendor market covers analytics and skills

Selected patterns, roles, and players, not exhaustive<sup>1</sup>

#### Operating-model archetypes

Where AI decisions and delivery sit; maturity moves from central to federated

- 1** Central CoE   **2** Hub-and-spoke   **3** Federated on a control plane

#### New roles

The role ladder that appears as agent counts grow

- O** Agent owner   **B** Orchestrator, "agent boss"   **F** Fleet manager

#### Adoption analytics and enablement

Connect usage data to productivity and targeted enablement

- WalkMe   Worklytics   Viva Insights   Pendo   DX

#### Skills and academies

Role-based AI skills at scale; best when fed by real workflow traces

- Sana   Workera   Internal academies   Multiverse   Section

<sup>1</sup>Logos are the property of their respective owners and appear for identification only.

Source: McKinsey, State of AI (November 2025)<sup>4</sup> and Superagency (January 2025)<sup>5</sup>; BCG, AI at Work (June 2025)<sup>11</sup>; Microsoft Work Trend Index (April 2025)<sup>15</sup>; NCSC (May 2026)<sup>28</sup>

| ARCHETYPE                   | VALUE PROPOSITION                                             | CHOOSE WHEN                                              |
|-----------------------------|---------------------------------------------------------------|----------------------------------------------------------|
| <b>Central CoE</b>          | Control and standards while capability is scarce              | Level 3; deliberately dissolve it into the hub over time |
| <b>Hub-and-spoke</b>        | Central platform and evals, federated build in the business   | Levels 4-5; the dominant mature pattern                  |
| <b>Trace-based coaching</b> | Enablement from real work, not courseware                     | From Level 4, once traces exist                          |
| <b>Census amnesty</b>       | Shadow builders become registered owners instead of offenders | The first people move, at any level                      |

#### WHAT GOOD LOOKS LIKE, BY LEVEL

| Level 1                | Level 2                    | Level 3                            | Level 4                             | Level 5                             |
|------------------------|----------------------------|------------------------------------|-------------------------------------|-------------------------------------|
| IT admin owns licenses | AI lead and CoE, champions | Named owner per workflow and agent | Fleet managers, pattern marketplace | Organization designed around fleets |

**DIAGNOSTIC** *Who is accountable, by name, if the flagship AI workflow fails? Are shadow builders drafted into ownership or treated as offenders?*

#### SO WHAT

Run the census with amnesty, register every agent with a named owner, and put your best shadow builders in charge of the patterns everyone else reuses.

# Five moves, in an order you cannot skip

**Dependencies, not ambition, dictate the sequence.** Attribution needs traces, traces need a chokepoint, safe autonomy needs identity and evals. Organizations that invert the order, expanding what agents may do before instrumenting what agents actually do, produce the incident record of 2025-2026. The five moves below cross the Level 3-4 boundary in dependency order; each produces a named artifact a board can inspect.

Exhibit 22

Five moves cross the chasm; each unlocks the next and leaves an inspectable artifact

The Level 3 to Level 4 crossing, in dependency order

|                                                                                                                                                                                                       |                                                                                                                                                                                                                                  |                                                                                                                                                                                                         |                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                             |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>1</b><br><b>Gateway and trace log</b><br>All model traffic through one chokepoint; raw keys move into it; every call traced with full payloads.<br><i>Artifact: the trace log · Owner: the CIO</i> | <b>2</b><br><b>Census and registry</b><br>One-week census with amnesty: logs, spend, self-registration. Every agent gets a named owner; unowned agents are switched off.<br><i>Artifact: the agent registry · Owner: the CIO</i> | <b>3</b><br><b>Attribution join</b><br>Token to team to use case to outcome, reported monthly to the CFO. Budgets and alerts per use case.<br><i>Artifact: cost-per-outcome report · Owner: the CFO</i> | <b>4</b><br><b>Identity and golden set</b><br>Per-agent credentials with scopes and a tested kill switch; a golden set gates every change to the flagship workflow.<br><i>Artifact: kill-switch test, eval run · Owner: the CISO</i> | <b>5</b><br><b>Gated autonomy</b><br>The flagship workflow runs end to end: reads free, writes gated, exceptions escalated. Autonomy widens as evals earn it.<br><i>Artifact: the instrumented workflow · Owner: the business-line lead</i> |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

Source: Agentic Stack Matrix, Hildesheim and Hainzl, August 2026; sequence validated against NCSC and Five Eyes deployment guidance<sup>28</sup>, Microsoft agentic adoption maturity model<sup>30</sup>, and production practice

The first 90 days

| DAYS 0-30 · SEE                                                                                                                                                                   | DAYS 31-60 · GOVERN                                                                                                                                                                                   | DAYS 61-90 · PROVE                                                                                                                                                                                       |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Census with amnesty</b> across logs, OAuth grants, and spend. <b>Gateway live</b> for the two teams with the largest model bills. First honest answer to "which agents exist". | <b>Registry with named owners;</b> unowned agents off. <b>Attribution join</b> for the top five use cases. <b>Per-agent credentials</b> on everything that writes; kill switch tested once, for real. | <b>One flagship workflow instrumented</b> end to end with gates and a golden set. <b>Cost per completed task</b> reported against the human baseline. Scale-or-stop decision on evidence, not anecdotes. |

**The prize is compounding, and it goes to whoever holds the record.** Instrumented operations turn AI from a story into a line item with unit economics: governed projects reach production at a multiple of ungoverned ones, specialist models cut unit costs by orders of magnitude on stable workflows, and each instrumented workflow makes the next one cheaper. The layer that sees the work becomes the layer that runs it. That is the strategic reason to cross now, at a moment when 95 percent of spend still produces no measurable return: the gap between the instrumented few and everyone else is the widest competitive opening this technology cycle has produced.

**Where to go from here.** Run the one-hour self-placement this week: nine rows, one artifact of evidence per cell, the median and the lagging rows on a single page. If you want the facilitated version with peer comparison, or you operate at Levels 4 to 6 and want to compare notes, write to [fabian.hildesheim@cdtm.com](mailto:fabian.hildesheim@cdtm.com).

# Sources

Numbered markers throughout the report point to the entries below. Live leaderboard figures carry their access date; every other figure was verified against the cited edition during August 2026. Where a link points to a domain rather than a document, the cited title locates the source on that site. Links were live as of August 10, 2026. Bundled entries group closely related sources, so one marker can repeat across pages.

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## About this report

**What it is.** This report is the diagnostic companion to the Agentic Adoption Levels (Levels 0-6), a staged framework for enterprise AI adoption from unaware to agent-operated. The matrix opens the single ladder into nine stack layers so that leadership teams can locate themselves per layer, read their profile, and execute upgrade moves in dependency order.

**Method.** Every statistic was verified against its primary source (report PDF, press release, or publisher page) in two research passes during August 2026; publication month and sample size are cited with each figure where available. Level descriptions are written as observable states with named artifacts, following behaviorally anchored assessment practice, because self-reported maturity inflates: the leading indices saw average scores fall as respondents learned what good looks like. The framework's design follows documented conventions: CMMI's capability profiles, SAE J3016's responsibility-defined levels, NIST CSF's current-versus-target profiles, and the CISA Zero Trust Maturity Model's pillars-by-stages structure. Numbered source markers throughout the report point to the references on the preceding page. The underlying timeline event list and level-distribution estimates are available from the authors on request.

**How to run the self-placement.** One hour, the leadership team and the operators who run the systems, one row at a time. Pick the cell that matches reality, name the artifact that proves it, and record disagreements between executives and operators; the disagreements are usually the finding. Output: a nine-row profile, the median level, and the two or three lagging rows whose upgrade moves become next quarter's plan. Reassess quarterly and expect early scores to drop before they rise.

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